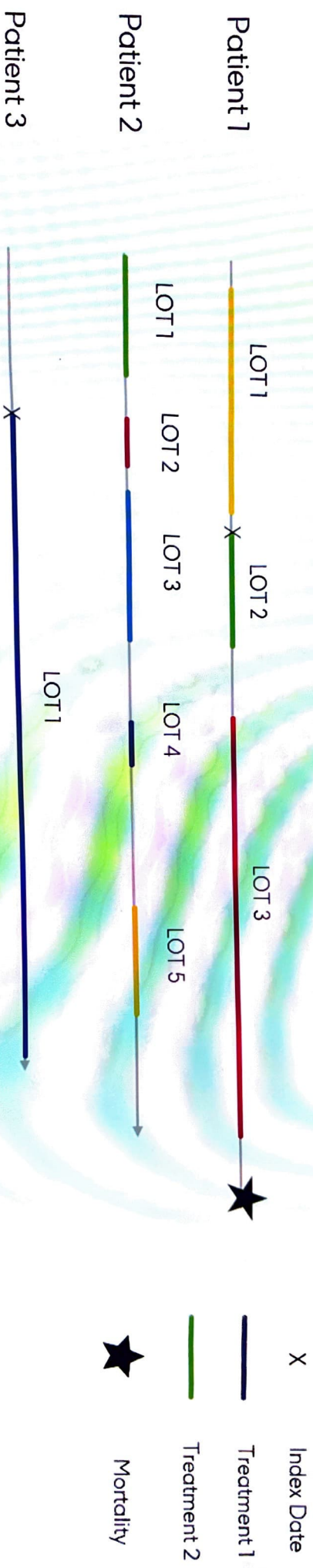


Flatiron NSCLC cohort extraction



Two cohorts generated:

- Nivolumab vs. Docetaxel – CHECKMATE057 trial
- Pembrolizumab + Chemo combo vs. Chemo combo – KEYNOTE407 trial definition

Cohort Extraction - Treatments

- 2025m06 datacut
- Users of a drug of interest is extracted by the 'lineoftherapy' table
 - 'linename' exactly match the treatment of interest
 - Multiple drug names in the line name indicate treatment combination
 - Other drugs might also be used within the same line of therapy and are reflected in the 'drugepisode' table, drugs not mentioned in the linename are ignored
 - Index date defined as the line start date. If there are discrepancy between 'lineoftherapy' and 'drugepisode' tables, use the date in the 'lineoftherapy' table
 - If the drug is used as the primary treatment in more than one line of therapy, the earliest line is used
 - All line numbers are included (not restricted to 1st or 2nd line)
- Remove patients who used both treatments of interest in any line of therapy

Cohort Extraction – Outcome

- Outcome: overall survival
- Flatiron: follow patients until death or end of follow-up period
- Extracted from 'enh_mortality_v2' table -> to be adjusted according to specs of oncology data product
 - Some patients' date of death is only available to year or month. If month information is not available, impute as '15' (June). If day information is not available, impute as '15'.
 - If the imputed date is earlier than the date of last visit, use the date of last visit instead.

Data Extraction

Variable	Table	Notes	To be adjusted
Mortality	enh_mortality_v2	- dateofdeath is imputed if it is only available to year or month. If month is missing, impute as '06' (June). If day is missing, impute as '15'. - If the imputed date is earlier than the last date of visit, change it to the last date of visit	- Y
Last date of visit	visit		
Start date	Lineoftherapy	- Use the linestartdate as the index date	
Line number	Lineoftherapy		
Baseline ECOG Score	Baseline_ecog	- Baseline ECOG are extracted for each line of therapy - If multiple baseline scores are extracted for the same line, use the one with test date closest to the index date	
Staging	enh_advancednsclc		
Histology	enh_advancednsclc		

Data Extraction

Variable	Table	Notes	To be adjusted
EGFR/ALK/KRAS/PD L1	Biomarker	- A patient is biomarker positive if the biomarker is tested positive at least once throughout the patient's record - Assume that biomarkers are time-invariant	- Y
Bilirubin/Hemoglobin/Platelets/Albumin/Lymphocytes/AST/Neutrophils/Creatinine/ALT/ALP	labs	- Use testbasename to match test type - Creatinine: use loinc code 2160-0 - Use the test result within [-45, 30] window of the index date. If multiple tests exist in the window, use the one closest to the index date.	- Y
Blood pressure	Vitals	- Use the test result within [-45, 30] window of the index date. If multiple tests exist in the window, use the one closest to the index date.	
Previous antineoplastic therapy	Lineoftherapy	- Linenumber == 1: no previous antineoplastic therapy - Linenumber > 1: has prior use of antineoplastic therapy	
Previous Platinum-based chemotherapy progression	Drugepisodes	- Progression information is incomplete in Flatiron - Use previous use of Pt-based chemotherapy as a proxy - Linenumber < linenumber of the drug of interest - Pt-based chemotherapy: cisplatin, carboplatin, and oxaliplatin	- Y
Previous EGFR inhibitor	Drugepisodes	- Erlotinib, Gefitinib, Afatinib, Osimertinib, Dacomitinib	- Y

Data Extraction

Variable	Table	Notes	To be adjusted
Previous use of CYP3A4 inhibitor	Drugepisodes	- Ketoconazole, itraconazole, ritonavir, clarithromycin, nefazodone	- Y
Previous anti-PD1 treatment	Drugepisodes	- Pembrolizumab, Nivolumab, Cemiplimab	- Y
Previous anti-PDL1 treatment	Drugepisodes	- Atezolizumab, Avelumab, Durvalumab	- Y
Previous anti-CDLA-4 treatment	Drugepisodes	- ipilimumab	- Y
Previous PD treatment	Derived	- Any previous anti-PD1 or anti-PDL1 or anti-CDLA-4 treatment	- Y
gender, race	demographics		
Smoking status	enh_advancednscic		
Age	Derived	- Derived as year of line start date – birthyear	
Maintenance therapy	Lineoftherapy	- Indicate whether the use of current therapy is maintenance	

Steps

Step 1: extract all qualifying participants and LOT info in R, intermediate dataset stored in Databricks [[link](#)]. No need to repeat.

Step 2: extract other features in Python
[query functions](#)
[prepare data table](#)

Ongoing work: adjust extraction details to be consistent with oncology data product specs